

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\max f(x) = 5x_1 + 4x_2 + 3x_3$$

$$-2x_1 - x_2 - x_3 \geq -5$$

$$-x_1 - 2x_2 - x_3 \geq -5$$

$$-x_1 - x_2 - 2x_3 \geq -5$$

$$x_1 + x_2 + x_3 \geq 2$$

$$x_1, x_2, x_3 \geq 0$$

$$x_1, x_2, x_3 \in \mathbb{Z}$$

Rilassamento Lineare

$$\max f(x) = 5x_1 + 4x_2 + 3x_3$$

$$-2x_1 - x_2 - x_3 \geq -5$$

$$-x_1 - 2x_2 - x_3 \geq -5$$

$$-x_1 - x_2 - 2x_3 \geq -5$$

$$x_1 + x_2 + x_3 \geq 2$$

$$x_1, x_2, x_3 \geq 0$$

Metodo delle due fasi

Fase I

$$\min f(a) = a_1$$

$$2x_1 + x_2 + x_3 + s_1 = 5$$

$$x_1 + 2x_2 + x_3 + s_2 = 5$$

$$x_1 + x_2 + 2x_3 + s_3 = 5$$

$$x_1 + x_2 + x_3 - s_4 + a_1 = 2$$

$$x_1, x_2, x_3, s_1, s_2, s_3, s_4, a_1 \geq 0$$

Calcolo dei costi

$$c_i = c_N - c_B^\top \cdot B_N^{-1}$$

Forma Standard

$$\min f(x) = -5x_1 - 4x_2 - 3x_3$$

$$-2x_1 - x_2 - x_3 - s_1 = -5$$

$$-x_1 - 2x_2 - x_3 - s_2 = -5$$

$$-x_1 - x_2 - 2x_3 - s_3 = -5$$

$$x_1 + x_2 + x_3 - s_4 = 2$$

$$x_1, x_2, x_3, s_1, s_2, s_3, s_4 \geq 0$$

$$c_1 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 0 - 1 = -1$$

$$c_2 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \end{pmatrix} = 0 - 1 = -1$$

$$c_3 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \\ 1 \end{pmatrix} = 0 - 1 = -1$$

$$c_4 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0 - 0 = 0$$

$$c_5 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = 0 - 0 = 0$$

$$c_6 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 0 - 0 = 0$$

$$c_7 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} = 0 + 1 = 1$$

$$c_8 = 1 - (0; 0; 0; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = 1 - 1 = 0$$

$$c_9 = 0 - (0; 0; 0; 1) \cdot \begin{pmatrix} 5 \\ 5 \\ 5 \\ 2 \end{pmatrix} = 0 - 2 = -2$$

Metodo del simplesso

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	a_1	b
s_1	2	1	1	1	0	0	0	0	5
s_2	1	2	1	0	1	0	0	0	5
s_3	1	1	2	0	0	1	0	0	5
a_1	1	1	1	0	0	0	-1	1	2
	-1	-1	-1	0	0	0	1	0	-2

Entra x_1 ed esce a_1 :

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	a_1	b
s_1	2	1	1	1	0	0	0	0	5
s_2	1	2	1	0	1	0	0	0	5
s_3	1	1	2	0	0	1	0	0	5
a_1	1	1	1	0	0	0	-1	1	2
	-1	-1	-1	0	0	0	1	0	-2

$$R1 \leftarrow R1 - R4 \cdot 2$$

$$R2 \leftarrow R2 - R4$$

$$R3 \leftarrow R3 - R4$$

$$R5 \leftarrow R5 + R4$$

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	a_1	b
s_1	0	-1	-1	1	0	0	2	-2	1
s_2	0	1	0	0	1	0	1	-1	3
s_3	0	0	1	0	0	1	1	-1	3
x_1	1	1	1	0	0	0	-1	1	2
	0	0	0	0	0	0	0	1	0

Fase II

$$\min f(x) = -5x_1 - 4x_2 - 3x_3$$

$$-x_2 - x_3 + s_1 + 2s_4 = 1$$

$$x_2 + s_2 + s_4 = 3$$

$$x_3 + s_3 + s_4 = 3$$

$$x_1 + x_2 + x_3 - s_4 = 2$$

$$x_1, x_2, x_3, s_1, s_2, s_3, s_4 \geq 0$$

Calcolo dei costi

$$c_1 = -5 - (0; 0; 0; -5) \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = -5 + 5 = 0$$

$$c_2 = -4 - (0; 0; 0; -5) \cdot \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix} = -4 + 5 = 1$$

$$c_3 = -3 - (0; 0; 0; -5) \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = -3 + 5 = 2$$

$$c_4 = 0 - (0; 0; 0; -5) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0 - 0 = 0$$

$$c_5 = 0 - (0; 0; 0; -5) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = 0 - 0 = 0$$

$$c_6 = 0 - (0; 0; 0; -5) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 0 - 0 = 0$$

$$c_7 = 0 - (0; 0; 0; -5) \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \\ -1 \end{pmatrix} = 0 - 5 = -5$$

$$c_8 = 0 - (0; 0; 0; -5) \cdot \begin{pmatrix} 1 \\ 3 \\ 3 \\ 2 \end{pmatrix} = 0 + 10 = 10$$

Metodo del simplesso

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
s_1	0	-1	-1	1	0	0	2	1
s_2	0	1	0	0	1	0	1	3
s_3	0	0	1	0	0	1	1	3
x_1	1	1	1	0	0	0	-1	2
	0	1	2	0	0	0	-5	10

Entra s_4 ed esce s_1 :

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
s_1	0	-1	-1	1	0	0	2	1
s_2	0	1	0	0	1	0	1	3
s_3	0	0	1	0	0	1	1	3
x_1	1	1	1	0	0	0	-1	2
	0	1	2	0	0	0	-5	10

$$R1 \leftarrow R1 \cdot \frac{1}{2}$$

$$R2 \leftarrow R2 - R1$$

$$R3 \leftarrow R3 - R1$$

$$R4 \leftarrow R4 + R1$$

$$R5 \leftarrow R5 + R1 \cdot 5$$

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
s_4	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	1	$\frac{1}{2}$
s_2	0	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	1	0	0	$\frac{5}{2}$
s_3	0	$\frac{1}{2}$	$\frac{3}{2}$	$-\frac{1}{2}$	0	1	0	$\frac{5}{2}$
x_1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	$\frac{5}{2}$
	0	$-\frac{3}{2}$	$-\frac{1}{2}$	$\frac{5}{2}$	0	0	0	$\frac{25}{2}$

Entra x_2 ed esce s_2 :

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
s_4	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	1	$\frac{1}{2}$
s_2	0	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	1	0	0	$\frac{5}{2}$
s_3	0	$\frac{1}{2}$	$\frac{3}{2}$	$-\frac{1}{2}$	0	1	0	$\frac{5}{2}$
x_1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	$\frac{5}{2}$
	0	$-\frac{3}{2}$	$-\frac{1}{2}$	$\frac{5}{2}$	0	0	0	$\frac{25}{2}$

$$R5 \leftarrow R5 + R2$$

$$R2 \leftarrow R2 \cdot \frac{2}{3}$$

$$R1 \leftarrow R1 + R2 \cdot \frac{1}{2}$$

$$R3 \leftarrow R3 - R2 \cdot \frac{1}{2}$$

$$R4 \leftarrow R4 + R2 \cdot \frac{1}{2}$$

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	b
s_4	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	1	$\frac{4}{3}$
x_2	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	0	0	$\frac{5}{3}$
s_3	0	0	$\frac{4}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	1	0	$\frac{5}{3}$
x_1	1	0	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0	0	$\frac{5}{3}$
	0	0	0	2	1	0	0	15

$$\begin{aligned}
 x_1 &= \frac{5}{3} \\
 x_2 &= \frac{5}{3} \\
 x_3 &= 0 \\
 s_1 &= 0 \\
 s_2 &= 0 \\
 s_3 &= \frac{5}{3} \\
 s_4 &= \frac{4}{3}
 \end{aligned}$$

La soluzione $x_{RL} = (\frac{5}{3}, \frac{5}{3}, 0)$ non è ammissibile per un problema di PLI.

Metodo Branch & Bound

Si prende la soluzione del Rilassamento Lineare, $\max f(x_{RL}) = 15$ e lo si pone come **Upper Bound**.

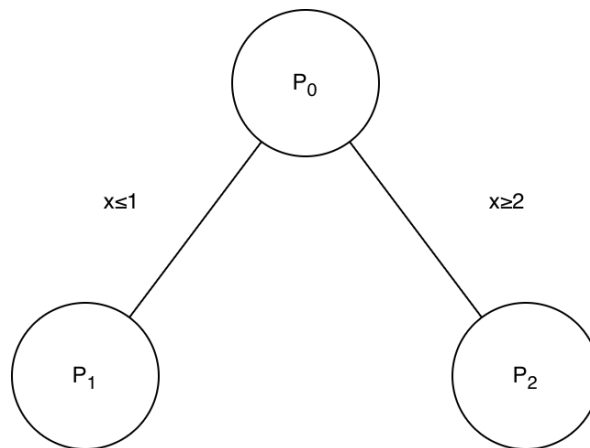
Si prende la prima componente frazionaria di x_{RL}

$$x_1 = \frac{5}{3}$$

Si ottengono i due sottoproblemi P_1 e P_2 da:

$$x_1 \leq \left\lfloor \frac{5}{3} \right\rfloor \vee x_2 \geq \left\lceil \frac{5}{3} \right\rceil$$

$$x_1 \leq 1 \vee x_2 \geq 2$$



P₁: $x \leq 1$

$$\begin{aligned}
 \max f(x) &= 5x_1 + 4x_2 + 3x_3 \\
 -2x_1 - x_2 - x_3 &\geq -5 \\
 -x_1 - 2x_2 - x_3 &\geq -5 \\
 -x_1 - x_2 - 2x_3 &\geq -5 \\
 x_1 + x_2 + x_3 &\geq 2 \\
 x_1 &\leq 1 \\
 x_1, x_2, x_3 &\geq 0
 \end{aligned}$$

Forma Standard

$$\begin{aligned}
 \min f(x) &= -5x_1 - 4x_2 - 3x_3 \\
 -2x_1 - x_2 - x_3 - s_1 &= -5 \\
 -x_1 - 2x_2 - x_3 - s_2 &= -5 \\
 -x_1 - x_2 - 2x_3 - s_3 &= -5 \\
 x_1 + x_2 + x_3 - s_4 &= 2 \\
 x_1 + s_5 &= 1 \\
 x_1, x_2, x_3, s_1, s_2, s_3, s_4, s_5 &\geq 0
 \end{aligned}$$

Metodo del simplesso duale

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	1	0	$\frac{4}{3}$
x_2	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	0	0	0	$\frac{5}{3}$
s_3	0	0	$\frac{4}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	1	0	0	$\frac{5}{3}$
x_1	1	0	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0	0	0	$\frac{5}{3}$
s_5	0	0	$-\frac{1}{3}$	$-\frac{2}{3}$	$\frac{1}{3}$	0	0	1	$-\frac{2}{3}$
	0	0	0	2	1	0	0	0	15

Entra x_3 ed esce s_5 :

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	1	0	$\frac{4}{3}$
x_2	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	0	0	0	$\frac{5}{3}$
s_3	0	0	$\frac{4}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	1	0	0	$\frac{5}{3}$
x_1	1	0	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0	0	0	$\frac{5}{3}$
s_5	0	0	$-\frac{1}{3}$	$-\frac{2}{3}$	$\frac{1}{3}$	0	0	1	$-\frac{2}{3}$
	0	0	0	2	1	0	0	0	15

$$R1 \leftarrow R1 + R5$$

$$R2 \leftarrow R2 + R5$$

$$R4 \leftarrow R4 + R5$$

$$R5 \leftarrow R5 \cdot -3$$

$$R3 \leftarrow R3 - R5 \cdot \frac{4}{3}$$

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	0	1	0	0	1	-1	2
x_2	0	1	0	-1	1	0	0	1	1
s_3	0	0	0	-3	1	1	0	4	-1
x_1	1	0	0	0	0	0	0	1	1
x_3	0	0	1	2	-1	0	0	-3	2
	0	0	0	2	1	0	0	0	15

Entra s_1 ed esce s_3 :

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	0	1	0	0	1	-1	2
x_2	0	1	0	-1	1	0	0	1	1
s_3	0	0	0	-3	1	1	0	4	-1
x_1	1	0	0	0	0	0	0	1	1
x_3	0	0	1	2	-1	0	0	-3	2
	0	0	0	2	1	0	0	0	15

$$\begin{aligned}
R5 &\leftarrow R5 \cdot -\frac{1}{3} \\
R1 &\leftarrow R1 - R3 \\
R2 &\leftarrow R2 + R3 \\
R5 &\leftarrow R5 - R3 \cdot 2 \\
R6 &\leftarrow R6 - R3 \cdot 2
\end{aligned}$$

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	1	$\frac{1}{3}$	$\frac{5}{3}$
x_2	0	1	0	0	$\frac{2}{3}$	$-\frac{1}{3}$	0	$-\frac{1}{3}$	$\frac{4}{3}$
s_1	0	0	0	1	$-\frac{1}{3}$	$-\frac{1}{3}$	0	$-\frac{4}{3}$	$\frac{1}{3}$
x_1	1	0	0	0	0	0	0	1	1
x_3	0	0	1	0	$-\frac{1}{3}$	$\frac{2}{3}$	0	$-\frac{1}{3}$	$\frac{4}{3}$
	0	0	0	0	$\frac{5}{3}$	$\frac{2}{3}$	0	$\frac{8}{3}$	$\frac{43}{3}$

$$\begin{aligned}
x_1 &= 1 \\
x_2 &= \frac{4}{3} \\
x_3 &= \frac{4}{3} \\
s_1 &= \frac{1}{3} \\
s_2 &= 0 \\
s_3 &= 0 \\
s_4 &= \frac{5}{3} \\
s_5 &= 0
\end{aligned}$$

La soluzione $x_{P_1} = (1, \frac{4}{3}, \frac{4}{3})$ non è ammissibile per un problema di PLI.

$$\mathbf{P_2}: x \geq 2$$

Forma Standard

$$\begin{aligned}
\max f(x) &= 5x_1 + 4x_2 + 3x_3 \\
-2x_1 - x_2 - x_3 &\geq -5 \\
-x_1 - 2x_2 - x_3 &\geq -5 \\
-x_1 - x_2 - 2x_3 &\geq -5 \\
x_1 + x_2 + x_3 &\geq 2 \\
x_1 &\geq 2 \\
x_1, x_2, x_3 &\geq 0
\end{aligned}$$

$$\begin{aligned}
\min f(x) &= -5x_1 - 4x_2 - 3x_3 \\
-2x_1 - x_2 - x_3 - s_1 &= -5 \\
-x_1 - 2x_2 - x_3 - s_2 &= -5 \\
-x_1 - x_2 - 2x_3 - s_3 &= -5 \\
x_1 + x_2 + x_3 - s_4 &= 2 \\
x_1 - s_5 &= 2 \\
x_1, x_2, x_3, s_1, s_2, s_3, s_4, s_5 &\geq 0
\end{aligned}$$

Metodo del simplesso duale

$$\begin{aligned}
\min f(x) &= -5x_1 - 4x_2 - 3x_3 \\
-2x_1 - x_2 - x_3 - s_1 &= -5 \\
-x_1 - 2x_2 - x_3 - s_2 &= -5 \\
-x_1 - x_2 - 2x_3 - s_3 &= -5 \\
x_1 + x_2 + x_3 - s_4 &= 2 \\
-x_1 + s_5 &= -2 \\
x_1, x_2, x_3, s_1, s_2, s_3, s_4, s_5 &\geq 0
\end{aligned}$$

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	1	0	$\frac{4}{3}$
x_2	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	0	0	0	$\frac{5}{3}$
s_3	0	0	$\frac{4}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	1	0	0	$\frac{5}{3}$
x_1	1	0	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0	0	0	$\frac{5}{3}$
s_5	0	0	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0	0	1	$-\frac{1}{3}$
	0	0	0	2	1	0	0	0	15

Entra s_2 ed esce s_5 :

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	1	0	$\frac{4}{3}$
x_2	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	0	0	0	$\frac{5}{3}$
s_3	0	0	$\frac{4}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	1	0	0	$\frac{5}{3}$
x_1	1	0	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0	0	0	$\frac{5}{3}$
s_5	0	0	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0	0	1	$-\frac{1}{3}$
	0	0	0	2	1	0	0	0	15

$$R1 \leftarrow R1 + R5$$

$$R3 \leftarrow R3 - R5$$

$$R4 \leftarrow R4 - R5$$

$$R5 \leftarrow R5 \cdot -3$$

$$R2 \leftarrow R2 - R5 \cdot \frac{2}{3}$$

$$R6 \leftarrow R6 - R5$$

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	s_5	b
s_4	0	0	0	1	0	0	1	1	1
x_2	0	1	1	1	0	0	0	2	1
s_3	0	0	1	-1	0	1	0	-1	2
x_1	1	0	0	0	0	0	0	-1	2
s_2	0	0	-1	-2	1	0	0	-3	1
	0	0	1	4	0	0	0	3	14

$$x_1 = 2$$

$$x_2 = 1$$

$$x_3 = 0$$

$$s_1 = 0$$

$$s_2 = 1$$

$$s_3 = 0$$

$$s_4 = 1$$

$$s_5 = 0$$

$$\max f(x) = 5x_1 + 4x_2 + 3x_3 \rightarrow 5(2) + 4(1) + 3(0) = 10 + 4 = \boxed{14}$$

Soluzione intera ammissibile trovata per il nodo P_2 .

$$\boxed{LB = 14 \quad UB = 15}$$